

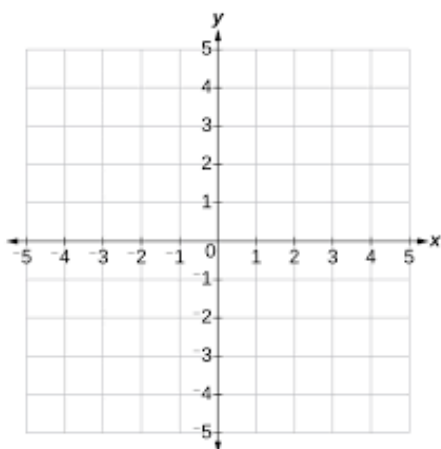
MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.1 Rectangular Coordinates in 3-Space, Spheres, & Cylindrical Surfaces

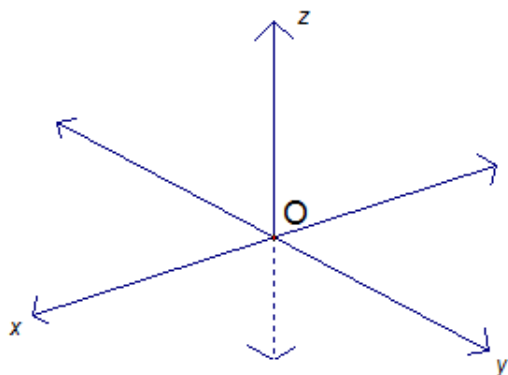
In calculus III one of our main focuses will be doing calculus in 3 dimensions. This is tremendously important in areas like physics, mechanical engineering, and civil engineering. So, MTH 202 will begin with the analytic geometry one must understand before doing calculus in three (and more than 3) dimensions.

Terminology:

- In the plane we draw an x and y axis and two numbers are required to locate a point in the plane.



- In the three dimensional rectangular coordinate system (also called 3 space or space) we add a z -axis and three numbers are required to locate a point in 3 space. A picture of the situation is below. In this class we will orient our axes as determined by the right-hand rule.

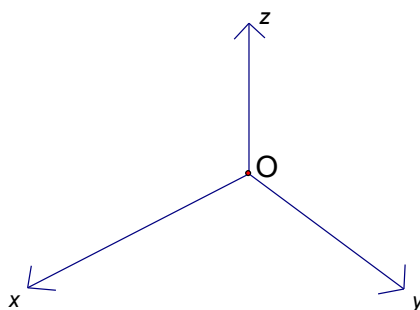


Note: The point labeled “O” is called the origin.

Note: In MTH 202 we will often represent three dimensional pictures on a two dimensional page. This presents challenges and forces us to think carefully about how we will draw figures. As the class goes on one should continue to practice three dimensional drawing.

Definition: The three coordinate axes determine three coordinate planes. Namely they determine the xy -plane, the yz -plane, and the xz -plane. These three coordinate planes divide space into eight parts, called octants. The first octant is determined by the positive axes.

The following is a picture of the first octant.

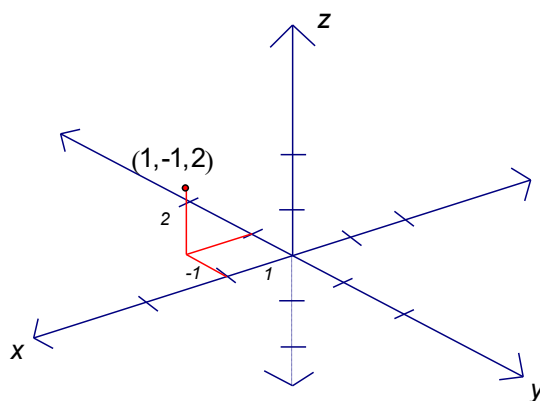


Notation: We represent a point in space by an ordered triple (a, b, c) of real numbers. We say a is the x -coordinate, b is the y -coordinate, and c is the z -coordinate.

Terminology: We let $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$ be the set of all ordered pairs of real numbers. In Calculus I and II you worked extensively with this set. We let $\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ be the set of all ordered triples of real numbers. In two dimensional analytic geometry, the graph of an equation involving x and y is a curve in $\mathbb{R}^2$. In three dimensional analytic geometry, the graph of an equation involving x , y , and z represents a surface in $\mathbb{R}^3$.

Example 1: Plot the point $(1, -1, 2)$ in $\mathbb{R}^3$.

The key to plotting a point in 3 space is creating part of a rectangular prism. As shown below (for more examples see the textbook).



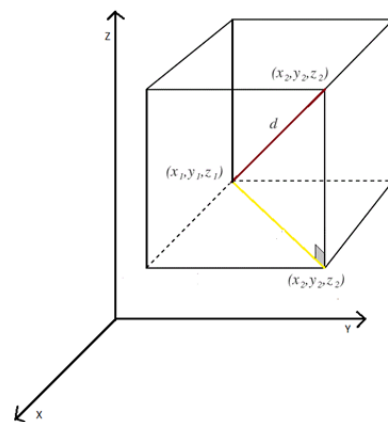
Theorem: Suppose that in space the coordinates of the point P_1 are (x_1, y_1, z_1) and the coordinates of the point P_2 are (x_2, y_2, z_2) . The distance between these points is written as $|P_1P_2|$ and is given by

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Proof:

Suppose you have a point P_1 with coordinates (x_1, y_1, z_1) and P_2 with coordinates (x_2, y_2, z_2) . We begin by assuming that $x_1 \neq x_2$, $y_1 \neq y_2$, and $z_1 \neq z_2$ otherwise can consider these points to exist in a 2-dimensional plane and the distance formula given will reduce to the distance formula in 2-dimensions.

We consider the cube formed by P_1 and P_2 . These points will form the corners of opposite sides of the diagonal of the cube. We then consider the point P_3 which has coordinates (x_2, y_2, z_1) .



Since P_1 and P_3 are in a plane together (which is parallel to the xy -plane) we can calculate the distance between them without considering their z -coordinate. We obtain:

$$d_{1,3} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Since P_2 and P_3 only differ in their z -coordinate the distance between them is formed by a straight line and we obtain:

$$d_{2,3} = (z_2 - z_1)$$

We then calculate the distance between P_1 and P_2 Using the Pythagorean theorem.

$$\begin{aligned} d_{1,2}^2 &= d_{1,3}^2 + d_{2,3}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \\ \Rightarrow d_{1,2} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \end{aligned}$$

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Question to the Class: How might one use the distance formula to derive an equation for a sphere with a center of (h, k, l) and a radius of r ?

Equation of a sphere: An equation of a sphere with center (h, k, l) and radius r is

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

Example 2: Find the center and radius of the sphere determined by the equation: $x^2 + y^2 + z^2 - 4z + 14y = -25$.

In order to solve this problem, we complete the square to get the equation in the desired form:

$$x^2 + y^2 + z^2 - 4z + 14y = -25$$

$$\Leftrightarrow x^2 + (y^2 + 14y + 49) + (z^2 - 4z + 4) = -25 + 49 + 4$$

$$\Leftrightarrow x^2 + (y + 7)^2 + (z - 2)^2 = 28$$

So, the equation determines a sphere with center $(0, -7, 2)$ and radius $2\sqrt{7}$.

Example 3: In no more than three sentences, describe in words what the region in $\mathbb{R}^3$ that is represented by the following inequalities: $4 < (x - 1)^2 + (y - 2)^2 + (z - 3)^2 < 9$ looks like.

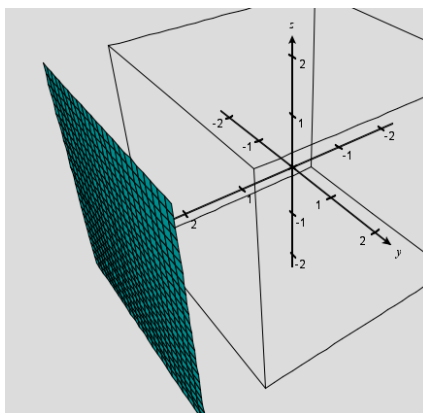
These inequalities represent the region that is outside of the sphere with radius 2 centered at $(1, 2, 3)$, and inside the sphere with radius 3 centered at $(1, 2, 3)$. The region contains no points which are on the surface of either sphere.

Cylindrical Surfaces

Definition: When any 2-dimensional curve is translated to create a 3-dimensional surface we call this surface a cylindrical surface.

Example 4: Plot the surface determined by the equation $x = 3$ in $\mathbb{R}^3$.

The key to this question is to first plot $x = 3$ in the xy -plane. Then, imagine what will happen as z varies. The result is below.

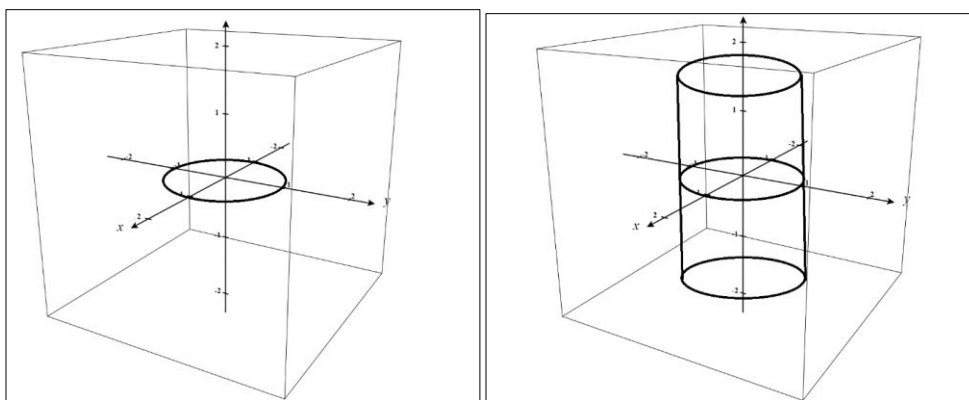


Note: If time permits, the instructor will introduce students to a free three-dimensional plotter which is available in Blackboard.

Example 5 (you try): Plot the surface determined by each of the following equations.

a) $x^2 + y^2 = 1$ in $\mathbb{R}^3$.

The graph should be an infinite right circular cylinder with a radius of 1.



b) $z = \sin(y)$ in $\mathbb{R}^3$.

The graph should be a sine wave in the yz -plane that is expanded to be an infinite wave in the x direction.

